

DELAWARE CITY, DE, USA

WMO: 997694

Lat: 39.582N

Lon: 75.589W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -7.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.1	(k) -1.4	(l) 10.5	(m) -1.3	(n) 4.3	(o) 300	(p) 0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.0	(c) 32.7	(d) N/A	(e) 31.4	(f) N/A	(g) 30.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.4	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.7	(b) 8.4	(c) 7.5	(d) DB	(e) -12.5	(f) 35.2	(g) 2.5	(h) 1.6	(i) -14.3	(j) 36.4	(k) -15.8	(l) 37.4	(m) -17.2	(n) 38.3	(o) -19.0	(p) 39.5	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.0	2.1	6.4	13.3	18.8	23.3	25.7	24.3	21.1	15.1	8.7	4.9
	DBStd	9.44	5.15	4.89	4.68	4.27	3.88	2.98	2.39	2.20	3.13	4.07	4.02	4.50
	HDD10.0	899	281	227	131	14	0	0	0	0	6	71	170	
	HDD18.3	2399	537	456	370	161	43	2	0	0	9	115	289	418
	CDD10.0	2283	2	5	19	113	273	400	486	444	332	166	32	11
	CDD18.3	739	0	0	0	9	57	152	227	186	92	16	0	0
	CDH23.3	5377	0	1	2	91	356	1097	2020	1282	466	63	0	0
	CDH26.7	1588	0	0	0	25	83	326	712	329	104	9	0	0
(14) Wind	WSAvg	3.1	3.3	3.5	3.8	3.6	2.9	2.8	2.4	2.4	2.8	3.1	3.2	3.2
(15) Precipitation	PrecAvg	1050	76	71	97	84	98	94	103	87	102	71	77	86
	PrecMax	1405	214	178	233	173	188	252	321	284	322	163	199	201
	PrecMin	632	13	11	21	9	6	5	23	6	21	2	12	20
	PrecStd	185	40	37	47	40	45	49	58	53	66	38	48	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	20.0	22.8	30.2	31.1	33.9	35.0	32.9	31.8	27.6	20.9	20.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	14.3	15.1	18.8	25.8	28.9	31.8	33.3	31.1	29.6	25.1	18.6	16.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	10.7	11.2	15.6	22.9	27.2	30.1	31.8	29.9	27.6	22.8	16.4	13.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	8.3	8.3	13.0	20.1	25.2	28.5	30.4	28.7	25.9	21.1	14.7	11.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	7.9	9.1	8.5	8.1	8.0	7.4	7.6	7.7	7.1	6.4
		MCDBR	11.5	11.7	12.5	13.1	10.7	9.8	9.7	9.0	9.4	9.6	9.6	9.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.316	0.322	0.365	0.403	0.448	0.511	0.521	0.515	0.426	0.364	0.328	0.312
	taud		2.446	2.420	2.356	2.277	2.186	2.053	2.038	2.040	2.265	2.443	2.501	2.503
	Ebn at Noon		852	899	888	873	837	782	769	761	814	839	832	831
	Edh at Noon		81	95	113	130	145	166	168	163	122	92	76	72
(44) All-Sky Solar Radiation	RadAvg		1.98	2.87	3.77	4.88	5.46	5.99	6.03	5.30	4.37	3.17	2.27	1.73
	RadStd		0.17	0.32	0.30	0.40	0.67	0.44	0.27	0.32	0.47	0.33	0.20	0.17

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (80 neighbors)		+0.46	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) +133	(l) +78

Nomenclature: See separate page